The $\mathrm{PSL}_3(4)$ incidence catalogue: classification, constructions and structural corrections

L100 source recovery

Abstract

We assemble a complete finite proof of the classification in the fixed 22-page Halle preprint of Gottschalk and Leemans. All subgroup anchors, primitive residue extensions and incidence isomorphism classes are covered. The resulting catalogue has 7, 7, 11 and 4 geometries in ranks 2, 3, 4 and 5, and none of higher rank. Literal projective constructions, doubling and truncation maps, residue data, stabilizer structures and complete automorphism and correlation groups are joined to the 29 printed rows. Several ancillary numerical, incidence and naming assertions require the explicit corrections below. The main classification is unchanged. Previously established classification inputs, algorithms and historical construction names are stated separately from the finite certificates.

1 Statement, conventions and fixed source

Let $G = \mathrm{PSL}_3(4)$ in its retained faithful 21-point realization, of order 20160. We consider finite coset geometries of rank at least two on which G acts flag-transitively. A geometry is *firm* if every panel has at least two objects, *residually connected* if every residue of rank at least two is connected, and *RWPRI* if the stabilizer of every flag acts primitively on at least one type of its positive-rank residue. The condition $(\mathrm{IP})_2$ says that every rank-two residue is a partial linear space or a generalized digon. No local two-transitivity condition is imposed in this classification.

We use right cosets with incidence given by nonempty intersection. Inversion converts the source's left-coset convention to this one. The standard firmness convention concerns panels, or equivalently nonmaximal flags in a geometry; the literal phrase "any flag" in the source definition must not be applied to a chamber itself. An isomorphism may permute entire types. The group $\mathrm{Aut}(\Gamma)$ fixes every type, whereas $\mathrm{Cor}(\Gamma)$ may permute them. An arbitrary automorphism of the uncolored incidence graph is not silently called a correlation.

Theorem 1. *Relative to the explicitly retained complete subgroup and exact computational inputs in Section 7, the firm, residually connected, G-flag-transitive RWPRI $(\mathrm{IP})_2$ geometries are precisely the 29 literal models in Table 1, up to incidence isomorphism with permutation of types. Their rank distribution is*

rank	2	3	4	5	≥ 6
number	7	7	11	4	0.

Every diagram, fusion count, structural group and construction clause has an explicit disposition below.

The source is the Halle report 99–30, *Geometries for the group* $\mathrm{PSL}(3, 4)$, by Harald Gottschalk and Dimitri Leemans [1]. Its exact 22-page file, page transcriptions and native diagrams are bound to the certificate. The 25-page journal edition has not been collated: a correction here is a statement about this fixed preprint. The original 55-clause reading register is preserved without alteration. The new register assigns every clause its proof, correction or retained antecedent; it does not overwrite the earlier reading-only record.

2 Complete primitive residue extension

The broad panel theorem [2] gives a finite reduction using primitivity alone. For a chamber stabilizer B and actual panel stabilizers $P_i > B$, put

$$K_\varnothing = B, \qquad K_S = \langle B, P_i : i \in S \rangle.$$

Recognition checks all intersections $K_S \cap K_T = K_{S \cap T}$, a maximal deletion $K_{S \setminus \{i\}} < K_S$ for each nonempty S, typed coset Helly for each subfamily of rank at least three, and the exact rank-two partial-linearity-or-digon test. The intersection and Helly conditions identify the residues with the corresponding coset geometries inside K_S. Maximality gives primitivity by the subgroup–block correspondence, even when the local action has a kernel. Connectedness gives generation of a residue stabilizer by its panel stabilizers. These observations establish both necessity and sufficiency, as proved in [2].

For completeness, take a target family of rank r. RWPRI supplies a maximal deletion H in its join K. By induction its rank-$(r-1)$ residue family has already occurred. The missing panel lies in K but outside H; adjoining it therefore generates K. All maximal overgroups and all eligible panels are tried. Stabilizer-orbit pruning uses actual automorphisms of the old tuple, and the final successive transporter/type-permutation test removes only equivalent families. No numerical signature is used as a sufficient isomorphism test.

The rank-two test retains every double coset. If $B = P \cap Q$ and $K = \langle P, Q \rangle$, the graph is a digon exactly when $|P||Q| = |K||B|$. Otherwise for each nontrivial BxB in $B \backslash Q / B$ the two neighbourhood sets

$$\{Qh : h \in P\}, \qquad \{Qhx : h \in P\}$$

must intersect in exactly one object. They already share Q; a second common object is precisely a quadrangle. This is the general test, not a shortcut valid only for two-transitive local actions.

The retained subgroup list has 95 G-conjugacy classes. Its permutation model is transported to the fixed realization by a checked bijective homomorphism with literal generator images. Complete local maximal inclusion data are transported by the same map. Every one of the 95 Borel anchors has now returned its final empty extension level. The final union has 145 full-group families up to inner conjugacy and type permutation: 21, 25, 63 and 36 in ranks 2–5. Earlier deadlines remain failures, not partial exhaustion certificates. The last anchor reused its completely saved levels and recomputed its unsaved next level from the beginning; its lost progress count was never treated as saved family data.

No higher family is omitted: every proper panel addition enlarges the join, so rank is bounded by the number of prime factors of $[G : B]$, with multiplicity. The acquired terminal empty-level checks, rather than agreement with the printed rank counts, establish exhaustion.

3 From specified actions to incidence classes

Inner equivalence does not identify all geometries. The complete automorphism group of G, of order 241920, is acquired with its actual generator-image maps. Its action and type permutations fuse the 145 families. The last 42 families were compared incrementally with the previous 103; previously completed comparisons were retained. The saved exact generator maps justify reuse of the already computed automorphism group and its cardinality. Computational faithfulness of the acquired nice representation is a stated input, not inferred from a cached size comparison.

Even full $\mathrm{Aut}(G)$ fusion alone would not classify bare incidence geometries: an incidence isomorphism need not normalize the chosen copy of G. The final comparison therefore uses the acquired palette construction [3]. Add one marker for every type, join it to exactly that type's objects, and distinguish object vertices from marker vertices by two colors. A color-preserving graph isomorphism restricts to precisely a type-permuting incidence isomorphism. Conversely

any such incidence isomorphism extends uniquely to the markers. Fixing the markers individually gives the type-preserving automorphism group. Thus the exact colored-graph computation supplies both bare equivalence and the full groups, without an uncolored shortcut.

All 29 resulting representatives have literal coset memberships, incidence matrices and generator actions. Each of the 145 families is joined to its representative; complete colored-graph comparisons separate the representatives. The rank distribution in Theorem 1 follows. The saved raw GAP output required a documented inverse of its line-continuation serialization; the derived files are bound to both the originals and 61 native round trips. A clean raw return alone is not used as evidence that malformed JSON was a scientific artifact.

4 The corrected catalogue

Table 1 uses the printed source-node order. Type sizes and panel orders therefore have an explicit node-by-node transport, including equal-sized types. The integer c is the number of inner conjugacy/type classes fused into the indicated incidence class; their sum is 145. The column $2T$ means that every panel action is two-transitive. The complete per-node flags are retained, including non-two-transitive primitive examples. Every bond, absent bond and printed ordered eccentricity/gonality triple is compared to the actual base residue, not merely to a diagram signature.

For the group columns put $A = \mathrm{Aut}(G)$ and let δ, σ, τ be diagonal, field and graph outer automorphisms. Their explicitly realized outer group is $S_3 \times C_2$, of order 12. Set

$$\Sigma = G \rtimes \langle \sigma \rangle, \quad T = G \rtimes \langle \tau \rangle, \quad U = G \rtimes \langle \sigma\tau \rangle,$$

$$\Gamma = G \rtimes \langle \delta, \sigma \rangle, \qquad V = G \rtimes \langle \sigma, \tau \rangle.$$

Here the notation denotes the actual extensions identified by the certificates, with the indicated outer images. Products $2 \times H$ in the table have a proved central complement, not just twice the order of H. The convention $\Sigma = P\Sigma L_3(4)$ is the extension by field automorphisms, as in the official Magma definition [11].

Table 1: All 29 corrected source rows.

Label	Type sizes	Panel orders	c	$2T$	Aut	Cor
2.1	$(21, 21)$	$(4, 4)$	1	yes	Γ	A
2.2	$(210, 56)$	$(14, 3)$	6	no	Σ	Σ
2.3	$(56, 280)$	$(1, 9)$	3	yes	V	V
2.4	$(120, 120)$	$(7, 7)$	3	yes	U	V
2.5	$(210, 120)$	$(6, 3)$	6	yes	Σ	Σ
2.6	$(280, 280)$	$(8, 8)$	1	yes	A	$2 \times A$
2.7	$(280, 1260)$	$(1, 8)$	1	yes	A	A
3.1	$(21, 210, 56)$	$(1, 4, 3)$	6	yes	Σ	Σ
3.2	$(56, 1260, 56)$	$(1, 8, 1)$	3	no	V	$2 \times V$
3.3	$(120, 210, 210)$	$(1, 2, 2)$	3	yes	Σ	V
3.4	$(56, 210, 210)$	$(1, 2, 2)$	3	yes	Σ	V
3.5	$(1120, 120, 120)$	$(6, 1, 1)$	3	no	U	V
3.6	$(56, 210, 56)$	$(3, 2, 3)$	6	yes	G	Σ
3.7	$(21, 105, 21)$	$(3, 4, 3)$	1	yes	Γ	A
4.1	$(21, 210, 56, 56)$	$(1, 1, 3, 3)$	6	yes	G	Σ
4.2	$(56, 1260, 21, 21)$	$(1, 4, 1, 1)$	3	no	Σ	V
4.3	$(120, 1260, 56, 56)$	$(1, 2, 1, 1)$	3	yes	Σ	V
4.4	$(21, 1260, 56, 56)$	$(1, 2, 1, 1)$	6	yes	Σ	Σ
4.5	$(56, 1260, 21, 21)$	$(1, 2, 1, 1)$	6	yes	Σ	Σ
4.6	$(120, 1260, 21, 56)$	$(1, 2, 1, 1)$	6	yes	Σ	Σ
4.7	$(120, 1260, 21, 21)$	$(1, 2, 1, 1)$	3	yes	Σ	V

Label	Type sizes	Panel orders	c	$2T$	Aut	Cor
4.8	$(56, 21, 21, 56)$	$(3, 1, 1, 3)$	6	yes	G	$2 \times \Sigma$
4.9	$(56, 21, 210, 56)$	$(1, 2, 2, 1)$	6	yes	G	Σ
4.10	$(21, 56, 56, 210)$	$(2, 1, 1, 2)$	12	yes	G	G
4.11	$(120, 120, 336, 336)$	$(1, 1, 2, 2)$	6	no	G	T
5.1	$(21, 56, 56, 210, 21)$	$(2, 1, 1, 1, 1)$	12	no	G	G
5.2	$(21, 56, 56, 210, 21)$	$(2, 1, 1, 1, 1)$	6	no	G	Σ
5.3	$(21, 56, 21, 56, 21)$	$(1, 1, 1, 1, 2)$	6	no	G	$2 \times \Sigma$
5.4	$(21, 56, 21, 56, 21)$	$(1, 1, 1, 1, 2)$	12	no	G	$2 \times G$

The exact structural table also records all 99 maximal parabolics and their 99 residue kernels, together with 29 Borels. Identical literal subsets reduce these positions to 77 atoms; 61 distinct actual parabolic/kernel pairs pay all quotient structures. The proof [9] uses faithful geometric actions, actual normal subgroups and complements, and complete normal forms. It never identifies a group solely from its cardinality. For example a hyperoval stabilizer has the entire alternating action on six points; a Baer stabilizer acts as the full seven-point Fano automorphism group; point and line stabilizers have actual translation radical 2^4 and Levi complement A_5. Small atoms are paid by cyclic, dihedral, quaternion and elementary-abelian semidirect normal forms.

Every full automorphism/correlation group is likewise identified inside the acquired complete palette group. Coordinate and duality actions are verified on every object and incidence, their actual outer images are recorded, and each extra central factor is split by an explicit binary adjustment of lifts [8]. Equality with the complete palette group uses exact orders after proving inclusion and the action map; equality of orders alone does not define the groups.

5 Constructions and their exact source joins

The two rank-five projective constructions use lines, points, two distinct hyperoval orbits and point pairs in $\mathrm{PG}(2, 4)$. The accepted coordinate-to-permutation map is checked on the actual generators. All ten pairwise incidence rules are tested, every partial flag extends to a chamber, and the chamber orbit has length 20160. Hence the action on chambers is regular, in particular kernel-free. Base-residue connectivity, primitivity and $(\mathrm{IP})_2$ checks transport to all residues by flag transitivity. All six ordered choices of distinct hyperoval classes have explicit coordinate/duality maps, checked on every relevant object type and incidence [4]. The single switched line–hyperoval rule gives Geometry 5.2.

For Geometry 5.1 the 20 extension numbers of Table 1 and all flag orders agree with the source. Lemmas 4.1–4.3 are paid by the complete flag, residue and action arguments. The supplement [10] identifies all 31 nonempty flag stabilizers, rather than inferring the intermediate names from those orders. In particular it proves $H_{123} \cong D_{10}$ and the actual fixed-point-free split $H_{25} \cong 2^4 : 3$. The source type order is fixed before these intersections are formed.

The Baer construction of 3.3 has 120 subplanes in a selected orbit, 210 point pairs and 210 line pairs. Its literal printed relation between the latter types needs the correction in Section 6. The corrected three-type incidence structure has explicit bijections to its classified coset representative; all incidences and actions are checked [5].

The six butterfly recipes 4.2–4.7 are supplied on all three Baer orbits, with 1260 ordered butterflies per orbit. Under the explicit all-four-cross-intersections convention, each positive pole has exactly one opposite pole. The stabilizer is proved to be $C_2 \times D_8$ by actual sixteen-word normal forms. The two hyperovals at each pole, their signs and central-line data are retained literally. Eighteen full object/incidence maps join these recipes to their six catalogue rows [7].

For 4.11 the two Baer classes are distinct. The relative oval and dual-oval orbit choice is explicitly supplied; it is not inferred from the printed clauses, which do not specify it. Six actual choices have complete four-type maps. Across these constructions the certificates check

all 1,935,360 coset members, 10,679,394 cross-type incidence pairs and 61,824 generator actions. These finite verification totals describe the supplied maps, not new classification counts.

Four doubles have complete literal isomorphisms:

$$2.7 \longmapsto 2.6, \qquad 4.1 \longmapsto 4.8, \qquad 5.2 \longmapsto 5.3, \qquad 5.1 \longmapsto 5.4.$$

The rank-two case has its own elementary incidence proof. The quoted general doubling antecedent requires rank at least three and its weaker $I_{1,2}$ shadow condition; neither hypothesis is silently changed. Sixteen new truncation maps and eight acquired maps pay the stated truncation links after the correction below. The literal four doubles, maps and fusion counts are in [6]. Earlier named constructions cited only by reference in the source remain attributed antecedents, as specified in Section 7.

6 Ancillary corrections and qualifications

These corrections leave the principal list of incidence geometries unchanged. Each is attached to a fixed source clause and exact literal countercheck, not to an uncollated journal edition.

1. In the diagrams 5.1 and 5.2 the 210-object node has panel order 1, not the printed 2. The complete construction flag counts and source-node transports agree on this correction.

2. The printed 3.6 truncation link from 5.3 is impossible: 5.3 has no type of 210 objects. The corrected 5.2 link has an explicit incidence isomorphism.

3. In the printed recipe for 3.3, the point-pair/line-pair relation has degree 18 and admits a flag with no extension to the Baer type. Requiring a common Baer subplane in the selected orbit gives degree 6 and the certified construction. This concerns the ancillary recipe, not nonexistence of Geometry 3.3.

4. Under the stated field convention for $P\Sigma L$, the typed group of 3.5 is U, and the correlation group of 4.11 is T, rather than the printed Σ. These are not relabelings based on orders. The three outer involution cosets contain 120, 280 and 336 involutions, respectively; adding the 315 inner involutions separates the three abstract extensions. Literal outer images and full group equalities are also certified.

5. In diagram 2.3 the actual kernel $3^2 : C_4$ in $3^2 : Q_8$ has index two and no involution outside it. Thus it has no complement. Under the usual split-colon convention the outer colon in $(3^2 : 4) : 2$ must be replaced by an extension dot.

Two further qualifications concern how a recipe is specified. On p. 4 the two displayed corresponding cross-intersections for butterflies alone do not force the advertised unique completion; explicit alternative completions are retained. The certified recipe imposes all four cross-intersections. This is a statement about the displayed conditions, not an adjudication of every external butterfly convention. On pp. 18–19 the oval/dual-oval relative orbit choice is supplied explicitly, as noted above.

The generic residue-kernel dot defined on p. 3 does not assert nonsplitting. Accordingly central 2.2^3 extensions are proved nonsplit separately, and unparenthesized $2^2 \times 2^2 : L$ displays are interpreted as the actual action on the entire radical; they do not imply an unproved central factor or invariant complement. The earlier working allegation of a missing horizontal bond in 3.5 was our own transcription error and is withdrawn. The native source has that bond; it is not among the source corrections.

7 Retained inputs, attribution and verification

The proof has the following explicit antecedent boundary.

- The complete TomLib subgroup classification for $L_3(4)$, its literal transported realization, and complete local maximal inclusion data are retained. GAP's exact subgroup, homomorphism, transporter and automorphism algorithms, and GRAPE/nauty's complete colored-graph computations are computational premises. Independent small-model checks and positive witness maps do not reprove these classification tables or software algorithms.

- The source's quoted Tits connectivity criterion and Buekenhout–Hermand coset-Helly criterion remain attributed with their hypotheses. The actual general recognition and augmentation arguments used here are the complete acquired proofs in [2]. We do not turn the source's short strongly-Boolean sentence into a stronger theorem lacking its setting.

- Classical terminology and uniqueness of $\mathrm{PG}(2,4)$ are retained. Every concrete point, line, hyperoval, Baer set and butterfly used by the current constructions is separately bound to its literal finite certificate. The general rank-at-least-three doubling theorem is retained exactly as quoted, with its $I_{1,2}$ condition; all four finite applications have direct maps.

- The source's retrospective thesis reduction and earlier Cayley/Magma computations are historical attribution. The present proof uses the complete acquired panel computation instead of claiming to rerun those programs. Earlier construction references, including the Gewirtz and unital graph names attached to 2.3 and 2.7, are retained as attributed antecedents. All 29 current coset models and all newly prescribed source recipes are supplied; we do not claim newly to inspect every cited primary article or recertify each external naming identification.

The machine-readable register preserves all 55 original clauses, 47 bibliographic attributions and the exact fixed-edition scope. All 29 diagram clauses acquire numerical, structural and construction routes, with the six correction records above (the two group-name sites are separate records). The source contains no named unresolved research question; omitted details and classification obligations are not relabeled as new questions. No present-day conjecture-status claim, Lean formalization, measurement or novelty claim is made.

The principal checker verifies every acquired input hash, the exact 95-anchor/145-family/29-model joins, all 55 source dispositions, the complete native manuscript inputs and damaged-interface controls. It does not repeat the acquired group or graph censuses. The complete proofs named below and their separate review records are included in the acquisition closure. Authorship of this synthesis and the component-specific review roles are recorded explicitly; root's previews are not substituted for final distinct acceptance.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* $\mathrm{PSL}(3,4)$, Halle Reports on Algebra and Geometry 99–30, fixed 22-page preprint.

[2] *Primitive residue extension for a fixed finite acting group*, acquired broad preparation v2; complete target, continuation and fusion certificates, L100 finite archive.

[3] *Literal palette isomorphisms and exact serialization repair*, acquired L100 incidence archive, v1.

[4] *The projective constructions for the* $\mathrm{PSL}_3(4)$ *catalogue*, acquired construction component v1.

[5] *The Baer-subplane construction and its printed-rule correction*, acquired L100 component v1.

[6] *Doubling, truncation and fusion joins*, acquired L100 component v1.

[7] *Ordered butterflies, ovals and their incidence constructions*, acquired L100 component v2.

[8] *Automorphism and correlation group structures in the* $\mathrm{PSL}_3(4)$ *catalogue*, acquired L100 component v1.

[9] *Parabolics, Borels and residue kernels in the* $\mathrm{PSL}_3(4)$ *catalogue*, acquired L100 component v2.

[10] *Intermediate flag stabilizers in Geometry 5.1*, acquired L100 supplement v1.

[11] Magma Handbook, projective classical groups, definition of $P\Sigma L$: `https://magma.maths.usyd.edu.au/magma/handbook/text/713`. Exact inspected notation record retained in [8].